

The importance of the early 1970s in Internet history

One of the major characteristics of the emerging network is innovation. One development quickly leads to another. In the early 1970s, scientists tried to overcome new problems. The new communication ideas, the experiments, the testing, and the tentative designs brought about an endless stream of networks that were ultimately interlinked to become the Internet. Someone had to record all the protocols, the identifiers, networks and addresses and the names of all the things in the networked universe. And someone had to keep track of all the information that stemmed from the discussions. That someone was Jonathan B. Postel, a young computer scientist who worked at that time on the ARPA project at UCLA (Cerf, 1998). Postel devoted himself to building and running the Internet's naming and numbering structure. He proposed the top-level domains dot-com, dot-edu, and dot-net (Hafner & Lyon, 1998). In those pioneering, unstructured and building years, Postel was, in effect, the Internet Assigned Numbers Authority (IANA). Postel was not elected to the position of responsibility he held in the Internet community; he was simply, in the words of the White House's Internet policy adviser, Ira Magaziner, "the guy they trust".

Secondly, the ARPANET succeeded in connecting the computers used in different time-sharing systems. Now they wished to connect the packet-switching network of the ARPANET with a satellite packet-switching network and a packet radio packet-switching network. In July 1970, the first packet radio with the concept of random packet transmission, ALOHANET, was developed at the University of Hawaii by Norman Abramson and became operational.

ALOHANET linked the University of Hawaii's seven campuses to each other and to the ARPANET. Based on this model, ARPA built its own packet radio network which was called PRNET. At that same period of time, ARPA also developed a satellite network, called SATNET.

In 1971, the UNIX operating system was developed at Bell Lab, quickly gaining the appreciation of many scientists. UNIX provides a suite of programs which makes the computer work. It is a stable, multi-user, multi-tasking system for servers, desktops and later on also for laptops. In 1972, ALOHANET connected to the ARPANET and a commercial version of ARPANET, called TELNET, became the first Public Packet Data Service. The Telnet protocol was a relatively simple procedure. It was a minimal mechanism that permitted basic communication between two host machines. Telnet applications allow users to log on and to operate remote computers. Such applications can, for example, be used to search and consult remote databases such as library catalogues.

In 1973, ARPANET was connected to international hosts. File Transfer Protocol (FTP) came into existence and worked using a Client Server Architecture. The file-transfer protocol specified the formatting for data files traded over the network. FTP made it possible to share files between machines. Moving files might seem simple, but the differences between machines made it very difficult. FTP was the first application to permit two computers to cooperate as peers instead of treating one as a terminal to the other. Telnet, FTP and TALK were the first applications to become available on ARPANET and are still used in some form or another on the Internet today. TALK was the first program that allowed Netusers to engage in a real-time conversation over the network. Netusers typed messages onto a split screen and they read replies at the bottom of the screen.

By early 1973, the network had grown to 35 nodes, and it had expanded to 38 host computers. Later that year, the network arrived in Norway and England and traffic was expanding significantly. In 1974, Vint Cerf and Robert Kahn developed a set of protocols that implemented the open architecture philosophy. These new protocols were the Transmission Control Protocol (TCP) and the Internet Protocol (IP).

TCP includes rules that computers on a network use to establish and break connections; IP includes rules for routing of individual data packets. The Transmission Control Protocol/Internet Protocol (TCP/IP) organizes the data into packages, puts them into the right order on arrival at their destination, and checks them for errors.

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